

Chapter 2.7: Related Rates

Word Problems: Tips

- Write down all important information and unknowns.
- Draw figures if you need them to help.
- Write down equations that relate to the quantities with regard to the related rate problems.
- Differentiate with respect to t and solve

Ex) The radius of a sphere is increasing at a rate of 2 mm/s. How fast is the volume increasing when the diameter is 80 mm?

* no not want V but $\frac{dV}{dt}$

* Since radius & Volume are increasing not constant

note: does not mean $r = 2 \text{ mm/s}$

$$\frac{dR}{dt} = 2 \text{ mm/s}$$

$$D = 80 \text{ mm}$$

$$r = 40 \text{ mm}$$

note! $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dt} = \frac{4\pi}{3} \frac{d}{dt}(r^3)$$

$$= \frac{4\pi}{3} \cdot 3r^2 \frac{dr}{dt}$$

$$= 4\pi \cdot (40)^2 \cdot 2 = 12800\pi$$

Ex) If a snowball melts so that its surface area decreases at a rate of 3 cm²/min, find the rate at which the diameter decreases when the diameter is 11 cm.

S - Not constant

$$\frac{dS}{dt} = -3 \text{ cm}^2/\text{min}$$

negative since decreasing $D = 11 \text{ cm} \rightarrow r = \frac{11}{2}$

$$\frac{dD}{dt} = \text{looking for}$$

note! S.A. of sphere. SA: $4\pi r^2$

$$S = 4\pi r^2$$

$$\frac{dS}{dt} = 4\pi \cdot 2r \frac{dr}{dt}$$

$$-3 = 4\pi \cdot 2\left(\frac{11}{2}\right) \frac{dr}{dt}$$

$$-3 = 44\pi \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{-3}{44\pi}$$

$$\therefore D = 2r$$

$$\frac{dD}{dt} = 2 \frac{dr}{dt}$$

$$\frac{dD}{dt} = 2 \left(\frac{-3}{44\pi} \right)$$

$$\frac{dD}{dt} = -\frac{3}{22\pi}$$

Ex) At noon, ship A is 170 km west of ship B. Ship A is sailing east at 40 km/h and ship B is sailing north at 35 km/h. How fast is the distance between the ships changing at 4:00 PM?

starting point



Note: distance formula

$$d = r \cdot t$$

want $\frac{dz}{dt}$

↳ 4 hrs have passed

$$\frac{dx}{dt} = 40 \text{ km/h}$$

$$\frac{dy}{dt} = 35 \text{ km/h}$$

$$D_A = 40(4)$$

$$D_B = (35)(4)$$

$$= 160 \text{ km}$$

$$= 140 \text{ km}$$

$$z^2 = x^2 + y^2$$

$$z = \sqrt{10^2 + 140^2}$$

$$= \sqrt{19700}$$

$$= 10\sqrt{197}$$

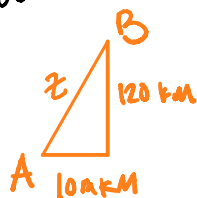
$$(z^2)' = (x^2)' + (y^2)'$$

$$2z z' = 2x x' + 2y y'$$

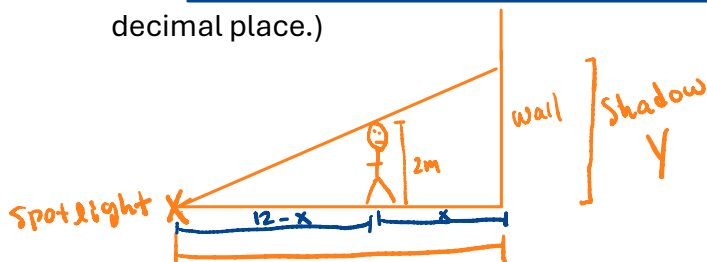
$$z' = \frac{2x x' + 2y y'}{2z}$$

$$z' = \frac{(10)(-40) + 140(35)}{10\sqrt{197}} = \frac{3200}{10\sqrt{197}} \text{ km/h}$$

finish

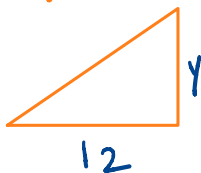
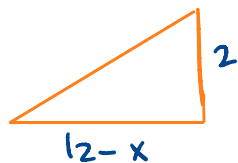


Ex) A spotlight on the ground shines on a wall 12 m away. If a man 2 m tall walks from the spotlight toward the building at a speed of 1.3 m/s, how fast is the length of his shadow on the building decreasing when he is 4 m from the building? (Round your answer to one decimal place.)



12 m

walking $\rightarrow 1.3 \text{ m/s}$



$$\frac{y}{12} = \frac{2}{12-x}$$

$$y = \frac{24}{12-x}$$

$$\frac{dx}{dt} = 1.3 \text{ m/s} \quad x = 4 \text{ m}$$

$$\frac{dy}{dt} = \text{want}$$

$$\frac{dy}{dt} = \frac{(12-x)(-24)' - 24(12-x)'}{(12-x)^2}$$

$$= \frac{-24x'}{(12-x)^2} = \frac{-24(1.3)}{8^2} = -0.5 \text{ m/s}$$